Version-Snapshot Stability of an AI-Presence Consistency Score

A pre-registered two-arm cross-generation test of the v1.7 CPC operationalization, with leave-one-provider-out sensitivity

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Working paper. Pre-registered at `v0.32-prereg-r2`
(design locked at `r1`*; acquisition prompt at* `r2`*). Prospective two-arm acquisition.*

Abstract

The AIAS™ (AI Availability Score) program operationalizes AI Availability as a measurable brand-growth layer alongside Mental and Physical Availability. Its Consistency component is presently scored by CPC = 1/(1+CV), the coefficient-of-variation transform locked in methodology version v1.7. v1.7 established that this score is not a mean-independent consistency construct — it is mechanically coupled to recall level ($|\rho|$ with Presence = 0.77) — and did not adopt it; a mean-independent redefinition was escalated to v1.8. The present study sets construct validity aside to ask a logically prior question: is the v1.7 score even version-stable? A measure that moves with the model vintage on which it is taken cannot track a brand across time, whatever its construct status. A pre-registered two-arm design measured CPC on a fixed 24-brand automotive registry under two model-version snapshots — an older vintage and the current frontier — across a six-model panel, holding probe wording, registry, and frame battery identical and varying only the dated model identifier; the contrast was made deliberately maximal across model generations. Brand-mention extraction reused the prior coder verbatim (pre-registered); a pre-scoring spot-check confirmed symmetric extraction across arms. Rank-order stability cleared the registered threshold marginally (Spearman ρ = 0.708) but proved fragile under a pre-registered leave-one-provider-out analysis (ρ ranging 0.48–0.82), with instability concentrated in the gpt-4o$\rightarrow$gpt-5.x jump. Magnitude stability was falsified (mean $|\Delta$CPC$|$ = 0.077, exceeding the 0.5·SD tolerance of 0.069), though with no net directional drift — version change reshuffles which brands read as consistent without shifting the overall level. The emerging-brand instability prediction was falsified: no brand crossed the recall floor between arms. The v1.7 CPC score is therefore at best partially and provider-dependently version-stable; version-fragility compounds the recall-coupling already identified in v1.7 as grounds for the v1.8 redefinition. The contribution is a characterization of the interim instrument, not a validation of it.

Keywords: AI availability; brand availability; AIAS; pre-registration; Ehrenberg-Bass; large language models; brand presence; measurement stability; model versioning; coefficient of variation

JEL codes: M31 (primary); L86; M37; D83

Paper status: Working paper. Pre-registered at `v0.32-prereg-r2` (design locked at `r1`; acquisition prompt at `r2`). Prospective two-arm acquisition.

1. Introduction

The AIAS™ (AI Availability Score) program treats AI Availability — the propensity of generative AI systems to surface a brand when buyers turn to them for category guidance — as a third measurable layer of brand availability alongside the Mental and Physical Availability of the Ehrenberg-Bass tradition (Sharp 2010; Romaniuk and Sharp 2022). The Presence component shipped first (AIAS™ 1.0; SSRN 6817841). The composite under construction adds further components, of which Consistency is the candidate for AIAS 2.0: the degree to which a brand's presence holds steady across the models constituting the measurement panel, rather than depending on which model is queried.

Consistency is presently operationalized as CPC = 1/(1+CV), where CV is the coefficient of variation of a brand's per-model recall count across the panel. Methodology version v1.7 (SSRN 6878818) subjected this operationalization to a discriminant-validity test and found it wanting: CPC is mechanically coupled to recall level — at recall counts in the 0–6 range the coefficient of variation tracks the mean by construction — so the score reparametrizes recall rather than isolating a mean-independent consistency dimension ($|\rho|$ with Presence = 0.77, above the program's 0.50 dissociation ceiling). v1.7 accordingly did not adopt CPC and escalated a mean-independent redefinition to methodology version v1.8.

That redefinition is the proper repair, and it is underway; this paper does not pre-empt it. It instead isolates a question that is logically prior to construct validity and that v1.8 must in any case answer: is the v1.7 score *version-stable*? A panel of large language models is not a fixed instrument. Its constituent models are deprecated and replaced on the providers' schedules — over the course of this work, the gpt-4o family was frozen while the frontier advanced to a separate gpt-5.x line. If a brand's CPC reading depends on the vintage of the panel on which it is taken, the score cannot be compared across time, and no construct-level repair in v1.8 rescues a quantity that is unstable to the panel's ordinary turnover. Version stability is thus a necessary, if not sufficient, property of any usable composite component, and characterizing it on the interim instrument is a direct input to the redefinition.

The study is a pre-registered, two-arm experiment. CPC is measured twice on a fixed automotive registry — once under an older model-version snapshot (Arm A) and once under the current frontier (Arm B) — across a six-slot panel, with probe wording, registry, cell structure, and frame battery held identical and only the dated model identifier varied between arms. The contrast was made deliberately maximal: each slot moves to its provider-tier's current production model — a cross-generation jump whose distance varies by provider, widest at the OpenAI gpt-4o$\rightarrow$gpt-5.x transition — so the design stresses version sensitivity rather than minimizing it. Three pre-registered hypotheses address rank-order stability (H_ScoreRankStable), magnitude stability (H_ScoreMagnitudeStable), and a directional prediction that thin-but-present emerging brands are the most version-sensitive subset (H_EmergingInstability). A pre-registered leave-one-provider-out analysis localizes any instability to particular providers, and a first-class flip-count diagnostic guards against rank statistics computed on a shrinking, survivor-biased set of brands.

The remainder of the paper proceeds as follows. Section 2 details the panel, registry, instrument, and the pre-registered scoring and falsification rules. Section 3 reports the three verdicts with their figures. Section

4 interprets a mixed result — marginal, provider-dependent rank stability against a falsified magnitude criterion — and its implications for v1.8 and the AIAS 2.0 composite. Section 5 states limitations, and Section 6 outlines the prospective tests the findings motivate.

2. Method

The design is a single-substrate, two-arm version contrast. A fixed automotive registry was scored twice under the v1.7 CPC instrument — once on an older model-version snapshot (Arm A) and once on the current frontier (Arm B) — with every element of the instrument held identical between arms and only the dated model identifier changed. Methodology, registry, panel, decision rules, and acquisition prompt were locked at git tags `v0.32-prereg-r1` (design) and `v0.32-prereg-r2` (acquisition prompt) before any data were collected.

The measurement panel holds six slots, two from each of three providers, fixed in composition since v0.17. For the version contrast each slot is instantiated at two dated snapshots: an older vintage and that provider-tier's current production model. Snapshots are pinned to dated model identifiers wherever the provider exposes them, so a call returns a fixed model rather than a drifting alias. The per-slot version distance is deliberately not uniform — it is whatever separates each tier's older and current production models — and it is widest for the OpenAI slots, where the gpt-4o family was frozen at its 2024 snapshot while the production frontier advanced to a separate gpt-5.x line.

Provider tier	Arm A (older snapshot)	Arm B (current)	Version distance
Anthropic flagship	claude-opus-4-5-20251101	claude-opus-4-8	within 4.x family (4.5 to 4.8)
Anthropic mid	claude-sonnet-4-5-20250929	claude-sonnet-4-6	within 4.x family (4.5 to 4.6)
OpenAI flagship	gpt-4o-2024-11-20	gpt-5.5-2026-04-23	cross-family (4o to 5.5), widest
OpenAI mini	gpt-4o-mini-2024-07-18	gpt-5.4-mini-2026-03-17	cross-family (4o to 5.4)
Google flash	gemini-2.5-flash	gemini-3.5-flash	one generation (2.5 to 3.5)
Google flash-lite	gemini-2.5-flash-lite	gemini-3.1-flash-lite	one generation (2.5 to 3.1)

The Google Arm-A slots are provider aliases rather than dated snapshots; no dated 2.5-class identifier is exposed, a limitation addressed in Section 5.

The substrate is automotive, with the 24-brand registry reused verbatim from AIAS v0.22 (SSRN 6829118), including its assignment of brands to four cells: heritage incumbents (Cell A), emerging disruptors (Cell B; Tesla, Rivian, Lucid, Polestar, Fisker), mass-legacy makes (Cell C), and defunct marques (Cell D; Pontiac, Oldsmobile, Plymouth, Mercury, Saturn). The cell structure is carried so version sensitivity can be examined where it is most plausible — the emerging disruptors, whose presence in model outputs is thin enough to move as training data advances.

Probe wording was reused verbatim from v0.22. Recognition (Phase A) poses one yes/no template per brand — whether the brand is commonly recognized as a car brand — across the 24 brands and six model slots, for 144 probes per arm. Recall (Phase B) is open-ended across a six-frame battery: three category frames (best brands, expert recommendations, highest quality and reliability) and three cultural frames (heritage and prestige, affluent and status-conscious choice, iconic and storied identity), posed once to each model, for 36 free-text responses per arm. Brands are not named in the recall prompts; they are extracted from the responses.

For each (brand, model) pair, the recall count $r \in \{0, \ldots, 6\}$ is the number of the six frames in which that model named the brand. Consistency follows the v1.7 definition: CPC = 1 / (1 + CV), where CV is the population coefficient of variation (standard deviation over mean, ddof = 0) of the six per-model recall counts, bounded in (0, 1] with higher values indicating more uniform recall across the panel. The v1.7 floor is inherited unchanged: a brand whose mean recall across the panel falls below 1.0 is recorded as undefined rather than assigned a value, a coefficient of variation being uninformative at near-zero signal. CPC is computed independently within each arm.

Brand mentions were detected with the v1.4 canonical coder reused verbatim — a word-boundary regular-expression match on the NFKD-normalized, lower-cased full registry name — applied identically to both arms; the verbatim reuse is the pre-registered element. Because a cross-version comparison is corrupted if the coder extracts asymmetrically across arms — were newer models to favor short forms (“Mercedes”, “VW”) that a full-name matcher misses — a pre-scoring spot-check verified symmetry before scoring. Across the abbreviation-prone brands, Arm B yielded marginally more full-name hits than Arm A (235 versus 220) at near-identical response lengths, and short-form-only occurrences the coder would miss totaled one across both arms. Extraction is therefore symmetric across the contrast, and the asymmetry that would have manufactured a spurious version effect does not arise. Per program discipline the coder was not modified to absorb the single short-form case; the verbatim instrument is retained and the case recorded.

Three hypotheses were pre-registered. H_ScoreRankStable (PRIMARY) holds that per-brand CPC preserves rank order across arms, confirmed if Spearman ρ(CPC_A, CPC_B) $\geq$ 0.70 over the pairwise-complete brands

— those with defined CPC in both arms — with pre-registered interpretive bands (rank-stable at $\rho \geq 0.70$, partially version-sensitive at 0.50–0.70, version-dominated below 0.50) and the verdict resting on the 0.70 line alone. H_ScoreMagnitudeStable (SECONDARY) holds that magnitudes are stable, the tolerance fixed scale-relative before data: mean $|\Delta\text{CPC}| \leq 0.5 \times$ SD(CPC_A), the standard deviation of Arm-A CPC over its defined brands, so the cut is anchored to the instrument's own between-brand spread rather than an absolute on a compressed scale. H_EmergingInstability (TERTIARY) predicts the Cell-B disruptors are the most version-sensitive subset, evaluated principally on the N/A$\rightarrow$defined flip rate — the live signal, since thin emerging brands are the candidates to cross the floor as models advance — and secondarily on the $|\Delta\text{CPC}|$ contrast among Cell-B brands defined in both arms; it is confirmed only if Cell B is the unique, non-zero highest-flip-rate cell.

Two pre-registered safeguards accompany the verdicts. A leave-one-provider-out analysis recomputes CPC over the four-model panel remaining after each provider's two slots are removed and re-derives ρ, localizing any rank instability to particular providers. And the count of brands flipping between defined and N/A across arms is reported as a first-class quantity alongside ρ, by direction and by cell, on the principle that flips are the most version-sensitive brands and excluding them from the pairwise-complete statistics would bias those statistics toward apparent stability; a high flip count qualifies any stability reading regardless of ρ. Falsification is by the registered lines: ρ < 0.70 for PRIMARY, mean $|\Delta\text{CPC}|$ > 0.5 · SD(CPC_A) for SECONDARY, and Cell B failing to be the unique non-zero highest-flip cell for TERTIARY.

3. Results

Acquisition completed without error — 144 recognition probes and 36 recall queries per arm, 360 calls in total, every response's provider-returned identifier matching its requested dated snapshot. Of the 24 brands, 14 carry a defined CPC in both arms and form the pairwise-complete set on which the rank and magnitude statistics are computed. The remaining ten fall below the recall floor in both arms — the four emerging disruptors (Rivian, Lucid, Polestar, Fisker), the mainstream incumbent Nissan, and the five defunct marques — and constitute an identical undefined set across the version contrast. No brand flips between defined and undefined in either direction. The flip-count guardrail is therefore quiet, and informatively so: the rank statistic is not computed on a set thinned by version-driven dropouts, so the stability findings below cannot be an artifact of survivor selection. That the undefined set is identical across a two-generation jump is itself the first result — the floor admits and excludes the same ten brands regardless of model vintage.

Rank-order stability (PRIMARY) is confirmed at the threshold but is provider-dependent. Spearman ρ between Arm-A and Arm-B CPC over the 14 pairwise-complete brands is 0.708, clearing the pre-registered 0.70 line and placing the result, by the registered verdict, in the rank-stable band (Figure 1). The margin is thin, and the pre-registered leave-one-provider-out analysis shows the confirmation is not robust. Removing the OpenAI slots *raises* ρ to 0.822; removing the Anthropic slots drops it to 0.512, and removing the Google slots to 0.482 — both into the partially-version-sensitive band. The rank stability that clears the threshold thus rests disproportionately on the two providers whose version jumps were smaller, while the OpenAI gpt-4o→gpt-5.x jump — the widest in the panel — is the principal source of rank disagreement, being the only provider whose removal improves the correlation. The defensible reading is not "CPC rank is version-stable" but "CPC rank is weakly and provider-dependently preserved, with the instability concentrated where the version distance is largest."

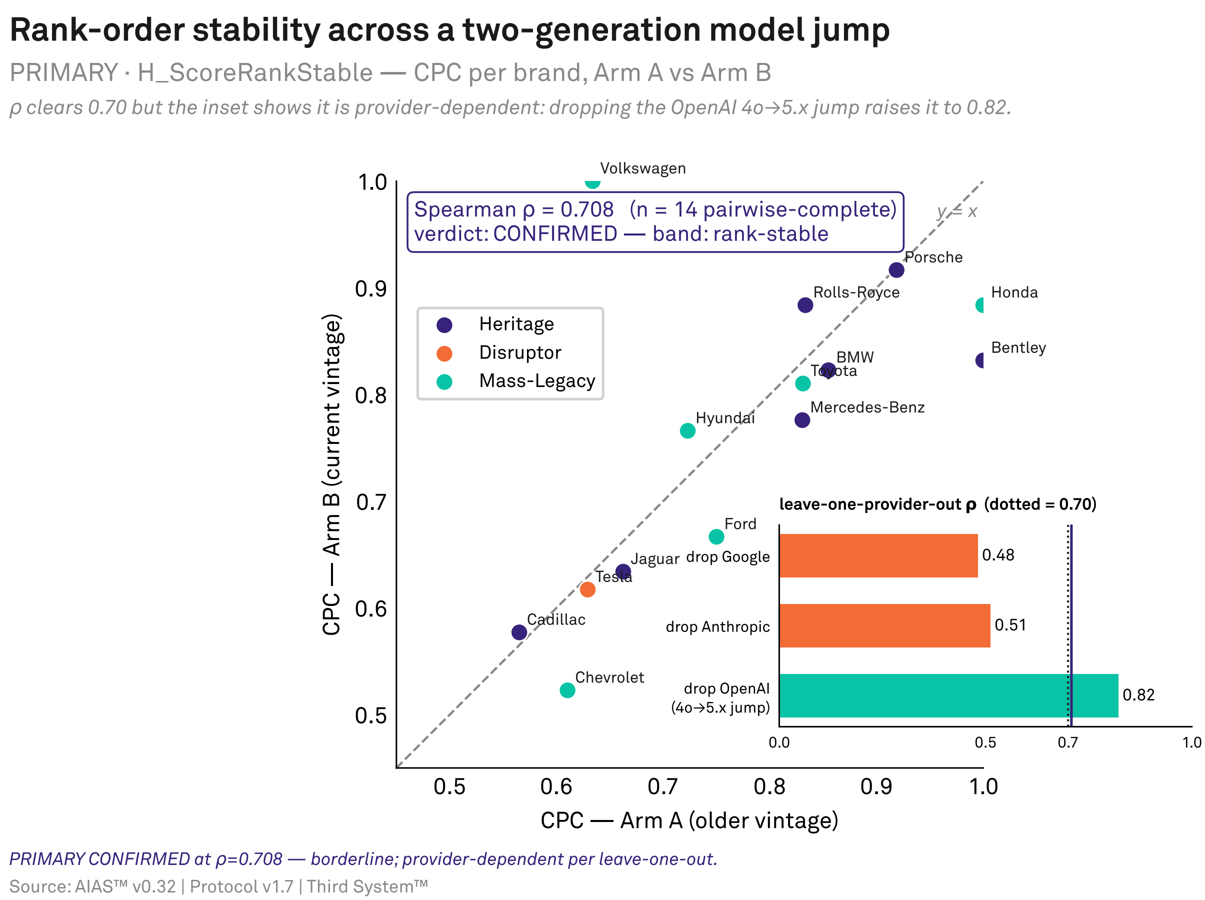

Figure 1: CPC per brand, Arm A against Arm B, over the 14 pairwise-complete brands; the diagonal is identity. The inset reports leave-one-provider-out Spearman ρ against the 0.70 threshold: the correlation rises when the OpenAI jump is removed and falls below threshold when either other provider is removed.

Magnitude stability (SECONDARY) is falsified. The mean absolute change in CPC across the pairwise-complete brands is 0.077, exceeding the pre-registered tolerance of 0.5 $\times$ SD(CPC_A) = 0.069 (Figure 2). The per-brand shifts are not, however, directional: the signed mean ΔCPC is -0.010, near zero, with shifts scattered on both sides. The version change does not move CPC up or down as a level; it reshuffles which brands read as consistent — some rise, some fall, by amounts that on average exceed half the between-brand spread the instrument uses to discriminate brands in the first place. Magnitude instability is therefore real but non-systematic: a brand's CPC value is not reproducible across the version contrast to within the instrument's own resolution, even as the panel-wide level is preserved.

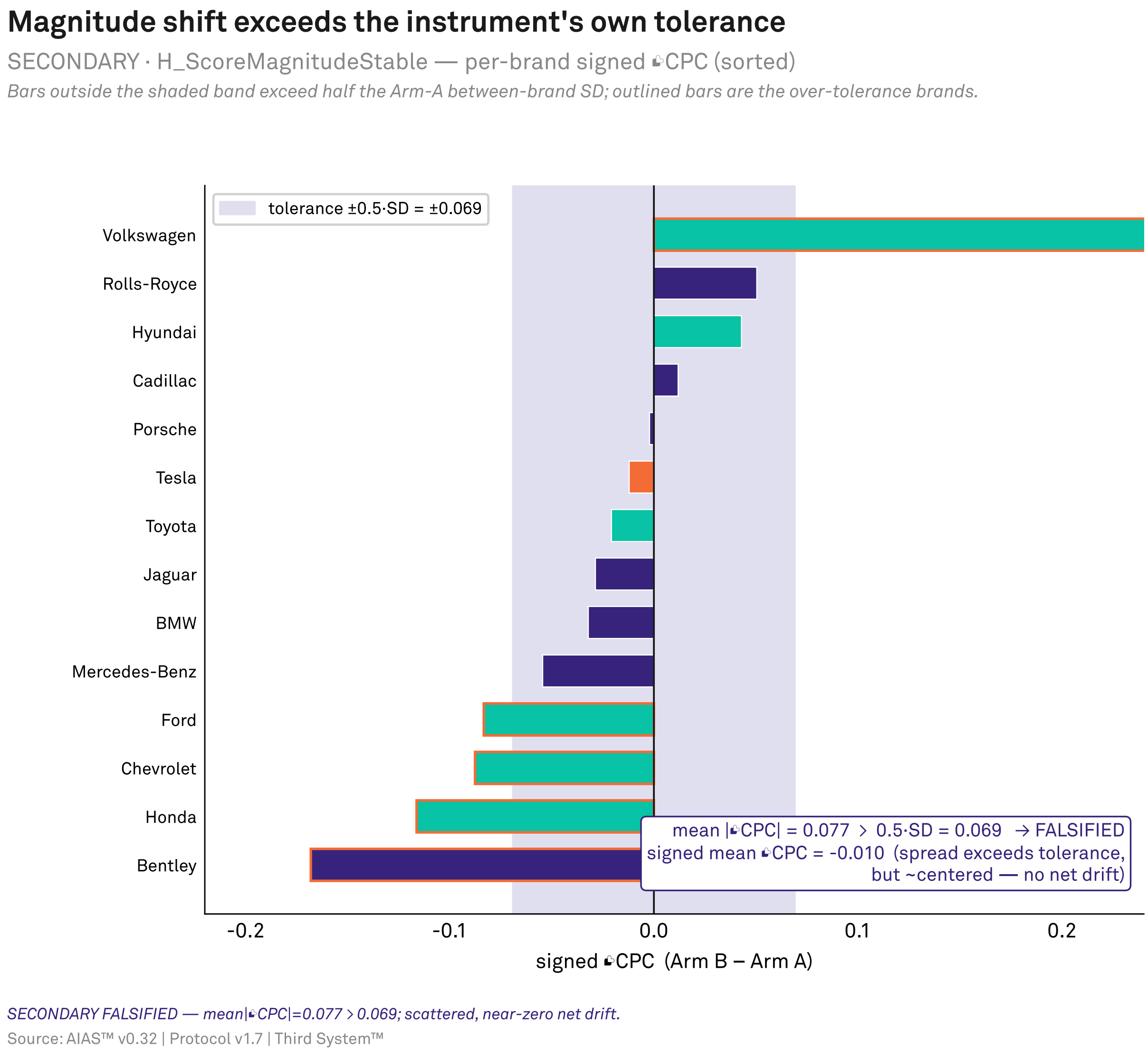

Figure 2: Signed ΔCPC (Arm B - Arm A) per pairwise-complete brand, sorted; the shaded band marks the ±0.5·SD(CPC_A) tolerance and outlined bars exceed it. The mean absolute shift (0.077) exceeds the tolerance (0.069) while the signed mean (-0.010) sits near zero.

Emerging-brand instability (TERTIARY) is falsified. The hypothesis predicted the Cell-B disruptors would be the most version-sensitive subset, surfacing as the highest N/A⟶defined flip rate as newer models learned thin-but-present brands. No brand crossed the floor in either direction in any cell; the flip rate is zero everywhere, so Cell B is not the unique non-zero highest-flip cell and the prediction is not observed (Figure

3, upper panel). The *direction* of the predicted effect is nonetheless visible below the floor, and is reported here as an explicitly exploratory, non-pre-registered observation. Among the disruptors, mean panel recall rose under Arm B — Rivian from 0 to 0.67, Lucid from 0 to 0.33 — as newer models began naming brands they had not named before, but the increases fall short of the mean-recall floor of 1.0, so no CPC becomes defined and no flip is recorded (Figure 3, lower panel). The binary floor thus masks a graded, direction-consistent version effect: emerging-brand presence is moving as predicted, but beneath the threshold at which the current instrument registers it. The observation motivates a prospective, pre-registered sub-floor analysis; it does not rehabilitate the falsified hypothesis.

Taken together, the three verdicts describe a score that is at best partially version-stable: rank order is preserved only weakly and only on the strength of the smaller-distance providers, magnitude is not preserved to the instrument's own resolution, and the version sensitivity the design was built to detect is present in the emerging subset but suppressed below the recall floor. Section 4 reads these results against the v1.7 construct-validity finding and the v1.8 redefinition.

4. Discussion

The three verdicts converge on a single reading: the v1.7 CPC score is, at best, partially and provider-dependently stable across model version. Rank order clears the pre-registered threshold but only marginally, and only on the strength of the providers whose version jumps were smallest; magnitude is not preserved to within the instrument's own discriminating resolution; and the version sensitivity the design was built to detect is present in the emerging subset but held below the recall floor. None of these is the clean stability a usable composite component would require. A measure intended to track a brand's cross-model consistency over time must return the same reading, to within its resolution, when the panel beneath it is refreshed; the v1.7 score does not.

The leave-one-provider-out result locates where that failure originates. The single provider whose removal *improves* the rank correlation is OpenAI, whose slots underwent the widest version jump in the panel — a full cross-family transition from gpt-4o to the gpt-5.x line — while the providers whose removal collapses the correlation, Anthropic and Google, moved within or by a single generation. Removing a provider also recomputes CPC over a smaller panel, so the analysis cannot isolate a provider's contribution with full purity; but the direction is unambiguous and consistent with the pre-registered expectation that rank instability concentrates where version distance is greatest. The implication for a continuously refreshed panel is direct: a composite whose components are upgraded on the providers' own cadences inherits the instability of whichever provider takes the largest generational step between readings, and averaging across the panel does not neutralize it. Stability is not a property the panel holds once and for all; it is contingent on how

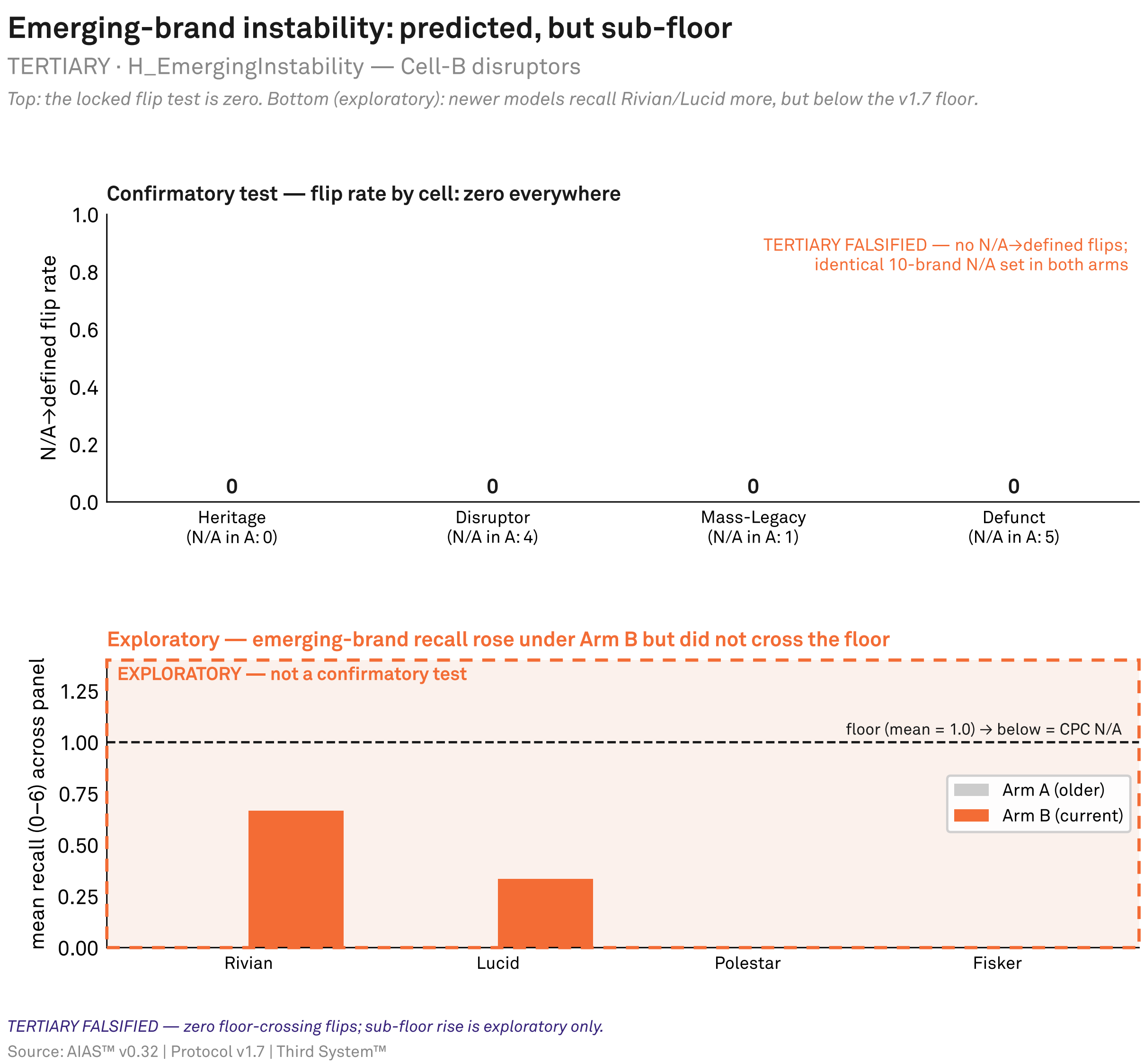

Figure 3: Upper panel (confirmatory): N/A⟶defined flip rate by cell, zero in every cell, with the identical ten-brand undefined set noted. Lower panel (exploratory, visually demarcated): mean panel recall for the Cell-B disruptors under each arm against the mean-recall floor of 1.0, showing recall rising under Arm B without crossing the floor.

far, and how unevenly, the constituent models move between measurements.

The magnitude result is in some respects the more telling of the two failures. The mean absolute shift exceeds the instrument's tolerance, but the signed mean is near zero: the version change does not raise or lower consistency as a level, it redistributes it across brands, some rising and some falling with no aggregate direction. Because the v1.7 score reparametrizes recall level, a uniform version-driven shift in how much the panel recalls would have appeared as net drift; its absence indicates that the instability lies less in the overall recall level than in which models surface which brands from one vintage to the next. The two failures the program has now documented in this score — its coupling to recall level (v1.7) and its fragility to version (this study) — are thus not obviously the same defect: the version-fragility survives even where the recall level is preserved, which suggests it is at least partly a genuine instability in the cross-model dispersion rather than a re-expression of the recall coupling. Disentangling level-driven from dispersion-driven version sensitivity is a question the redefinition must take up rather than assume away.

The recall floor shapes what the score can see, and the composition of the undefined set is itself informative. The same ten brands fall below the floor in both arms, in three kinds. The defunct marques are correctly absent — no version of any model recalls a discontinued brand, and stability there is trivial. The emerging disruptors are absent for a different reason: they are present and rising — recall for Rivian and Lucid increased under the current panel — but beneath the threshold at which a coefficient of variation becomes informative, so the instrument records them as unchanged while their presence is in fact moving. The third kind is the most pointed: Nissan, a mainstream global incumbent, sits below the recall floor in both arms. Market dominance and recall salience do not coincide, echoing the decoupling observed in premium spirits, where category-leading houses likewise went unrecalled in the operative discourse (v0.23, SSRN 6834298). The floor filters the registry not by market presence but by discursive salience in the recall frames, and CPC — defined only above that floor — is a consistency reading over the discursively salient subset, not the registry as a whole. Whatever Consistency becomes under v1.8, the boundary of where it is defined will carry this property forward.

For the redefinition and for the AIAS 2.0 composite, the contribution is twofold. First, version-fragility joins recall-coupling as a second, at-least-partly-independent reason the v1.7 score cannot serve as the Consistency component: v1.8 must clear not only the mean-independence bar v1.7 set but a version-robustness bar this study sets, and against a controlled cross-version contrast rather than a single panel snapshot. Second, the no-net-drift finding constrains what a version-robust measure looks like: because the instability is a brand-level redistribution rather than a level shift, a redefinition that merely normalizes recall level will not on its own deliver version stability; the measure must be robust to the cross-model surfacing pattern changing between vintages. Operationally, the same result tells the program that any longitudinal AIAS reading must control panel vintage — pinning dated snapshots and migrating them deliberately — or accept that a

Consistency reading is comparable across time only to within the wobble the most volatile provider introduces. The study does not validate the interim instrument; it characterizes precisely how, and where, that instrument fails to hold still, and hands the redefinition a concrete target.

5. Limitations

The most consequential limitation is the instrument itself. The score under test is the v1.7 operationalization, which v1.7 established is coupled to recall level rather than isolating a mean-independent consistency dimension. The version-stability findings are therefore findings about that recall-coupled score: the PRIMARY and SECONDARY tests are, by construction, tests of whether a recall-level-laden quantity holds its rank and magnitude across model version, not tests of a validated consistency construct. They characterize the interim instrument the program is replacing, and they do not transfer automatically to whatever mean-independent measure v1.8 adopts — that measure will require its own version-stability test. The contribution is scoped accordingly: a characterization of the score in use, not a general claim about the version stability of Consistency.

The empirical base is narrow in three respects that bound generalization. It is a single substrate: automotive was chosen for its mix of heritage, emerging, mass-legacy, and defunct brands, but consistency may behave differently in categories with denser or sparser discourse, and the cross-category variation the program has documented elsewhere cautions against reading one substrate as the whole. It is a single cross-version contrast: the design deliberately maximized the version distance, so the result describes the score under a large, asymmetric generational jump, not under the smaller increments a panel might more often see — a minimal-distance contrast could plausibly show more stability and was not tested. And it is a single wave: the two arms are measured at one point in time, so the study speaks to version stability — the same brands under different model vintages — but not to temporal stability, the same panel re-queried later. A further wrinkle follows from the program's history: because the panel was specified by floating aliases never pinned to dated snapshots, the older arm reconstructs the prior instrument only approximately, using the oldest still-served dated snapshots rather than the exact models earlier phases queried.

Provenance is incomplete on the Google slots. No dated identifier for the 2.5-class Gemini models is exposed, so the older Gemini arm is pinned to a provider alias rather than a frozen snapshot, and the API returned that alias as its own version metadata, leaving the exact underlying model unverifiable. The alias was the best-available provenance and is recorded as such; the risk that it was silently repointed between specification and acquisition was mitigated by acquiring promptly, but it cannot be ruled out from the returned metadata alone.

The statistical base is modest, and the modesty interacts with the design. Fourteen brands carry defined

CPC in both arms, so the rank correlation rests on a small sample; the wide swing under leave-one-provider-out is partly a small-n phenomenon as well as a substantive provider effect, and the confidence interval on a Spearman ρ of 0.708 at n = 14 is correspondingly broad. The TERTIARY test is thinner still: of the five Cell-B disruptors, only Tesla carries defined CPC in both arms, so the $|\Delta\text{CPC}|$ leg of the emerging-instability hypothesis was effectively untestable and the hypothesis rested on the flip rate alone — which was zero. Finally, the hard recall floor that renders CPC undefined below a mean of one mention cannot register sub-threshold movement, so the direction-consistent recall growth among the disruptors that the exploratory analysis surfaced is invisible to the confirmatory instrument by construction: a graded version effect can be present and unmeasured.

6. Future Research

The most direct successor is to re-run this design on the v1.8 measure once it is defined. The two-arm cross-version contrast generalizes beyond the present instrument: any candidate Consistency operationalization can be scored on a fixed registry under an older and a current panel snapshot and held to the same rank, magnitude, and flip criteria. Promoting that procedure to a standing requirement — a version-stability test each composite component must pass before adoption, on the same footing as the discriminant-validity test v1.7 introduced — would make version-robustness a gate rather than an afterthought, and would answer the question Section 5 leaves open: whether a mean-independent redefinition inherits the version-fragility documented here or escapes it.

Two refinements would deepen the measurement itself. The graded recall growth the exploratory analysis surfaced beneath the floor argues for a pre-registered floor-sensitivity analysis — measuring the sub-threshold recall movement directly, rather than only the binary crossing — so that a version effect present but unregistered by the current floor can be detected and quantified rather than inferred. And version distance, treated here as a fixed maximal contrast, should be treated as a designed variable: a minimal-distance contrast to complement this maximal one, and ideally a graded series of distances per provider, would convert the leave-one-provider-out observation — that instability concentrates where the jump is widest — from a single-point inference into a tested dose-response between version distance and measurement instability.

Two extensions would establish how far the result reaches. Replication across substrates would test whether the partial, provider-dependent stability seen in automotive holds in categories with denser or sparser discourse, where the floor admits different proportions of the registry. And a second measurement wave, re-querying a fixed panel vintage at a later date, would separate version change from calendar time — distinguishing instability caused by upgrading the models from instability that would arise even on an unchanged panel — which the single-wave design here cannot.

Finally, the result carries an operational recommendation for the program independent of any redefinition. Because the instability concentrates in the provider taking the largest generational step, and because providers deprecate and replace models on their own schedules — the gpt-4o freeze and the approaching 2.5-class Gemini retirements being the immediate cases — longitudinal AIAS measurement should migrate its panel deliberately rather than passively: pinning dated snapshots, re-snapshotting on a fixed cadence ahead of deprecation cliffs, and recording the panel vintage with every reading, so a change in a brand's score can be attributed to the brand rather than to the silent turnover of the instrument beneath it.

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Declarations

Conflict of interest. The author is employed by Samsung Electronics America in a corporate brand-creative and governance role. Samsung participates in the automotive sector as a component supplier rather than as a vehicle marque — through Harman International (in-vehicle audio and infotainment), Samsung SDI (battery cells), and Samsung Display (automotive displays) — but does not compete as an automotive brand, and no brand in the v0.22 registry scored here is Samsung-owned or -affiliated; a pre-acquisition conflict-of-interest screen confirmed this. The brand-level supply relationships underlying this disclosure are itemized in the conflict-of-interest statement of v0.22 (SSRN 6829118), cited above as the registry source. The research is conducted independently through Third System and is self-funded; Samsung Electronics America had no role in the study's design, data, analysis, or reporting. A single pre-registered instrument was applied uniformly across all brands in both arms, and no brand was singled out for differential treatment.

Funding. Self-funded.

Ethics. Not applicable; no human subjects; public APIs and LLM prompts only.

Data and pre-registration. Pre-registration, locked registry, two-arm acquisition data (Phase A and Phase B, both arms, provider-returned version metadata preserved), scoring code, and figures are deposited at the Open Science Framework (osf.io/ec6wh, v32 component). Methodology was locked at git tags v0.32-prereg-r1 (design) and v0.32-prereg-r2 (acquisition prompt) before any data were collected.